Year 7 Set 3 · White Rose · Lesson 2

Place value and ordering decimals

w/c 2 November 2026 · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Know the value of each digit in integers and decimals

2. Compare and order integers and decimals

Do now: retrieval practice

4 mins

1. Solve x + 9 = 20

   x = 11

2. Work out 3.5 + 2.7

   6.2

3. Write 4 thousands, 3 tens and 5 ones as a number

   4 035

4. Work out 7 × 8

   56

Objective 1

Know the value of each digit in integers and decimals

Key idea

Learn

!

Each column is worth 10 times the column to its right. After the decimal point come tenths, hundredths and thousandths. Multiplying by 10 moves the digits one place left; dividing by 10 moves them one place right.

Objective 1: Know the value of each digit in integers and decimals

I do: example 1

I do

What is the value of the 6 in 2 461?

1. Columns: 2 thousands, 4 hundreds, 6 tens, 1 one
2. The 6 is in the tens column
3. 6 tens
4. The value is 60

I do: example 2

I do

What is the value of the 7 in 3.472?

1. Columns: 3 ones . 4 tenths, 7 hundredths, 2 thousandths
2. The 7 is in the hundredths column
3. 7 hundredths = 7/100
4. The value is 0.07

I do: example 3

I do

Work out 4.3 × 10 and 56 ÷ 100

1. × 10: digits move 1 place left
2. 4.3 × 10 = 43
3. ÷ 100: digits move 2 places right
4. 56 ÷ 100 = 0.56

Mini whiteboards

We do

1. Value of the 5 in 4.153?

Answer: 0.05 (5 hundredths)

2. Work out 0.62 × 100

Answer: 62

3. Value of the 4 in 7.042?

Answer: 0.04 (4 hundredths)

4. Work out 3.6 ÷ 10

Answer: 0.36

You do: choose your challenge

You do

Bronze

Value of the 8 in 5.81?

Silver

Work out 2.7 × 100

Gold

Work out 45 ÷ 1000

Platinum

Which is larger: 0.909 or 0.99?

You do: answers

Check

Bronze	8 tenths (0.8)
Silver	270
Gold	0.045
Platinum	0.99

If you finish early

Extra

Bronze

Value of the 2 in 3 285?

200

Silver

Work out 0.09 × 1000

90

Gold

Work out 7 ÷ 100

0.07

Platinum

What do you multiply 0.045 by to get 45?

1000

Exam-style question 1

1 mark

Write down the value of the 9 in 15.293

Space for working

Mark scheme: 9 hundredths or 0.09 (B1)

Exam-style question 2

1 mark

Work out 3.08×1000

Space for working

Mark scheme: 3080 (B1)

Objective 1 achieved

Know the value of each digit in integers and decimals



Objective 2

Compare and order integers and decimals

✓ Achieved: Know the value of each digit in integers and decimals

Key idea

Learn

!

Line up the decimal points. Fill empty places with zeros so every number has the same number of decimal places. Compare from the left, one column at a time.

Objective 2: Compare and order integers and decimals

I do: example 1

I do

Which is larger, 2.09 or 2.1?

1. Write 2.1 as 2.10
2. Compare 2.09 and 2.10
3. Tenths: 0 is less than 1
4. 2.1 is larger

I do: example 2

I do

Order 0.6, 0.06, 0.66, 0.606, smallest first

1. Write each with 3 dp: 0.600, 0.060, 0.660, 0.606
2. Compare: 600, 60, 660, 606 thousandths
3. Order: 60, 600, 606, 660
4. 0.06, 0.6, 0.606, 0.66

I do: example 3

I do

Order −3, 2.5, −0.5, 0, smallest first

1. Negative numbers are less than zero
2. −3 is further below zero than −0.5
3. Then 0, then 2.5
4. −3, −0.5, 0, 2.5

Mini whiteboards

We do

1. Which is smaller: 0.45 or 0.405?

Answer: 0.405

2. Order 3.2, 3.02, 3.22, smallest first

Answer: 3.02, 3.2, 3.22

3. Order 0.8, 0.08, 0.88

Answer: 0.08, 0.8, 0.88

4. Which is larger: −1.5 or −1.2?

Answer: −1.2

You do: choose your challenge

You do

Bronze

Order 1.5, 1.05, 1.55, smallest first

Silver

Order −2, −2.5, 1, 0.5, smallest first

Gold

Write < or >: 0.38 _ 0.4

Platinum

Order 3/4, 0.7, 0.705, smallest first

You do: answers

Check

Bronze	1.05, 1.5, 1.55
Silver	−2.5, −2, 0.5, 1
Gold	0.38 < 0.4
Platinum	0.7, 0.705, 3/4 (0.75)

If you finish early

Extra

Bronze

Which is larger: 5.3 or 5.29?

5.3

Silver

Order 2.4, 2.04, 2.44, 2.404

2.04, 2.4, 2.404, 2.44

Gold

Order −0.3, −3, 0.03, 0.3

−3, −0.3, 0.03, 0.3

Platinum

Find a decimal halfway between 1.2 and 1.3

1.25

Exam-style question 1

2 marks

Write these in order, smallest first: 0.7, 0.07, 0.77, 0.707

Space for working

Mark scheme: 0.07, 0.7, 0.707, 0.77 (B2; B1 for three in order)

Exam-style question 2

1 mark

Write < or > between 4.06 and 4.6

Space for working

Mark scheme: 4.06 < 4.6 (B1)

Objective 2 achieved

Compare and order integers and decimals



Real life

Apply

Four runners' times (seconds): 12.4, 12.04, 12.44, 12.14. Who came first?

The smallest time wins: 12.04 s

Extension: stretch yourself

Stretch

1. Find a number between 0.35 and 0.36

e.g. 0.355

2. Work out 0.3×0.2

0.06

Exit ticket

Exit

1. What is the value of the 3 in 6.234? **(1 mark)**

2. Work out 0.52 × 100 **(1 mark)**

3. Write in order, smallest first: 0.5, 0.05, 0.55 **(1 mark)**

Exit ticket: mark scheme

Mark it

Q1 3 hundredths (0.03) [1]

Q2 52 [1]

Q3 0.05, 0.5, 0.55 [1]

Total: ____ / 3

Today I achieved

- ✔ **Know the value of each digit in integers and decimals**
- ✔ **Compare and order integers and decimals**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework

Due next lesson

1. What is the value of the 7 in 3.472? [1]
2. Put in order, smallest first: 0.6, 0.06, 0.66, 0.606 [2]
3. Work out 4.3 × 10 [1]
4. Work out 56 ÷ 100 [1]
5. Which is larger, 2.09 or 2.1? Explain. [1]

Extra practice: Corbettmaths → search "Ordering decimals"